SDM COLLEGE OF ENGINEERING AND TECHNOLOGY, DHARWAD.

AI-Based Automated Waste Segregation System

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➢ Problem Statement:

Improper waste segregation leads to environmental pollution, inefficient recycling, and increased landfill usage. Manual segregation is time-consuming, inconsistent, and unhygienic, especially at the primary collection level. There is a need for a smart, low-cost, and automated system that can accurately classify and segregate waste in real time with minimal human intervention using AI techniques.

➢ Proposed Solution:

1. Conveyor belt carries mixed waste
2. Camera captures waste images
3. Raspberry Pi processes and identifies material type
4. System classifies waste (plastic, metal, wood, cloth etc.,)
5. Motor driver controls sorting mechanism
6. Waste is directed into separate bins automatically
7. Reduces manual effort and improves segregation efficiency

Literature Survey:

Methodology	Technique used	Limitations	Accuracy	Reference
Manual segregation	Human sorting	Time-consuming, unsafe, error prone	~50-60%	Tchobanoglous et al., Integrated Solid Waste Management.
Sensor based system	Inductive, capacitive, moisture sensors	Limited material detection	~60-70%	Tchobanoglous et al., Solid waste engineering
Image processing	Digital Image Processing(color, shape, edge detection)	Sensitive to lighting, overlapping issues	~70-80%	Gonzalez & Woods, Digital Image Processing
Deep Learning	CNN, YOLO	High computation, large dataset required	~90-95%	J. Redmon et al., YOLO: “Real-Time Object Detection”, 2016.
Proposed system	YOLO+TensorFlow lite	Optimized for embedded constraints	~90%	TensorFlow Lite

Design Concept:

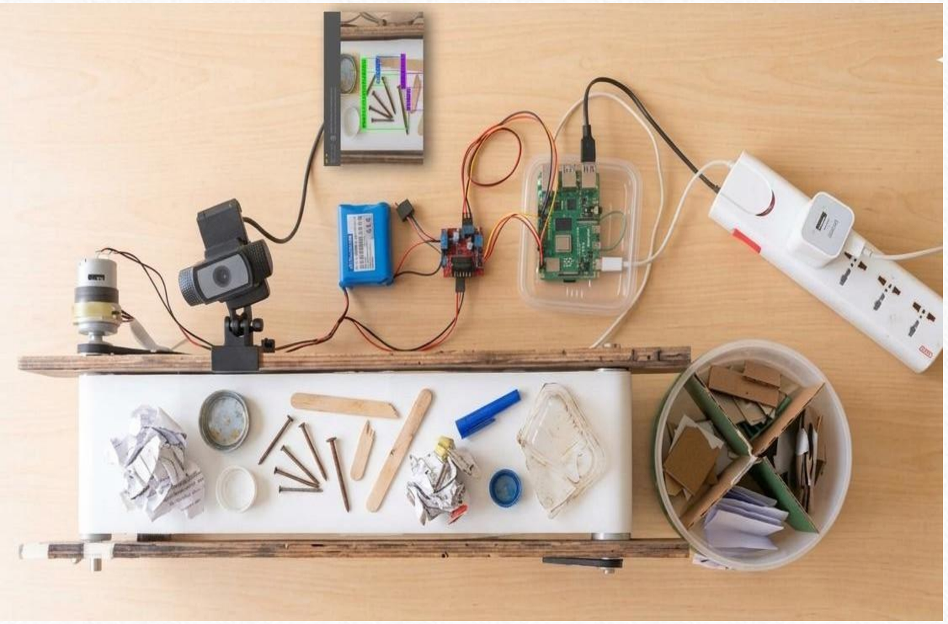


Figure-1: Model Setup

- The proposed design concept focuses on developing an automated waste segregation system that integrates mechanical movement, image processing, and embedded control into a single efficient unit.
- A conveyor belt continuously transports waste to a camera-based detection module, where visual data is analyzed to identify material types such as plastic, metal, or wood.
- This information is processed by the Raspberry pi which then sends control signals to a motor driver to actuate sorting mechanisms.
- The system is designed with a modular structure, ensuring easy scalability, low maintenance and adaptability for different waste categories.

Components Required

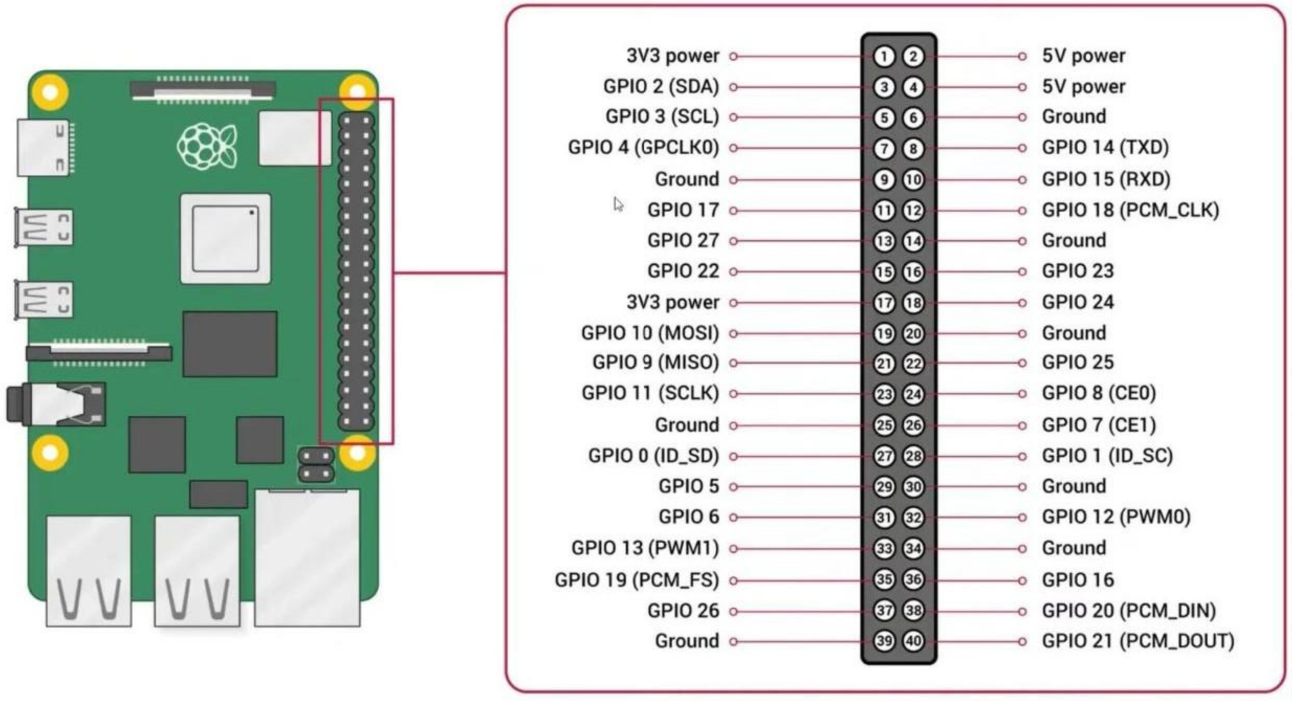
Hardware Components	Software Tools
Raspberry Pi 4 Controller board	Raspberry Pi OS
W200 Camera Module USB type	Python Programming Language
L298N Motor Driver	OpenCV
Servo Motor	Roboflow
12V DC Motor	TensorFlow
Custom built Conveyor belt	

Libraries Used

Library	Description
OpenCV (cv2)	Used for image capture and basic image processing from camera
NumPy (numpy)	Handles array operations and mathematical computations
TensorFlow	Runs the trained model for waste classification
Rpi.GPIO	Controls servo motors and hardware pins
Time module	Manages delays and timing for system operations

Hardware Components:

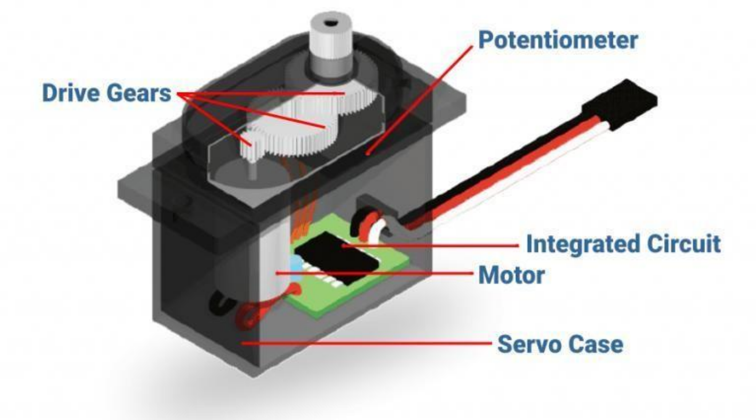
1. Raspberry Pi-



Raspberry Pi Features:

- Raspberry Pi 4 Model B
- Processor: 64-bit quad-core ARM Cortex-A72 (ARM v8)
- RAM: 4GB LPDDR4
- 2 micro-HDMI ports
- 2 USB 3.0 Ports
- 2 USB 2.0 Ports
- Gigabit ethernet port
- 802.11b/ g/n/ac wireless (Built in Wi-Fi module supporting both 2.4GHz and 5GHz bands.
- Bluetooth 5.0
- PoE (Power over ethernet, requires PoE HAT, sold separately)
- Power supply- 5V/3A USB-C

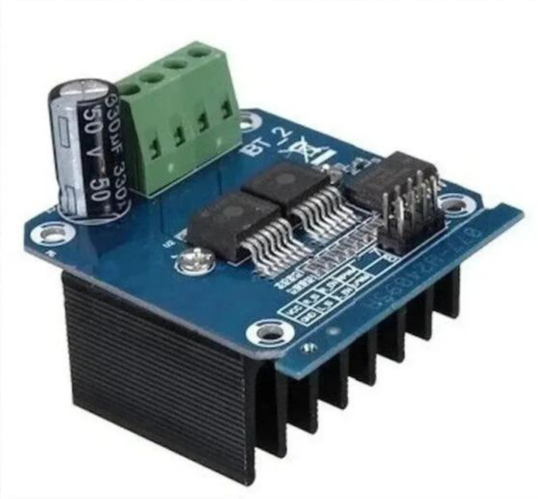
2. Servo Motor-



Servo Motor Features:

- This is a servo motor, commonly used for precise position control.
- It consists of a DC motor, gears, control circuit, and potentiometer.
- Drive gears reduce speed and increase torque for accurate movement.
- The potentiometer provides feedback about the shaft position.
- The integrated circuit (IC) controls motor rotation based on input signals.

3. Motor Driver-



Motor Driver BTS7960Features:

- Motor Driver IC: BTS7960 High Current Half-Bridge Driver
- Driver Type: H-Bridge Motor Driver Module
- Operating Voltage: 5V to 27V DC
- Logic Input Voltage: 3.3V – 5V compatible
- Maximum Current Capacity: 43A (peak)
- Continuous Current: Around 15A–20A with proper cooling
- PWM Frequency Support: Up to 25 kHz
- Control Method: PWM speed control + direction control

4. DC Motor: RS-555

DC Motor Features:

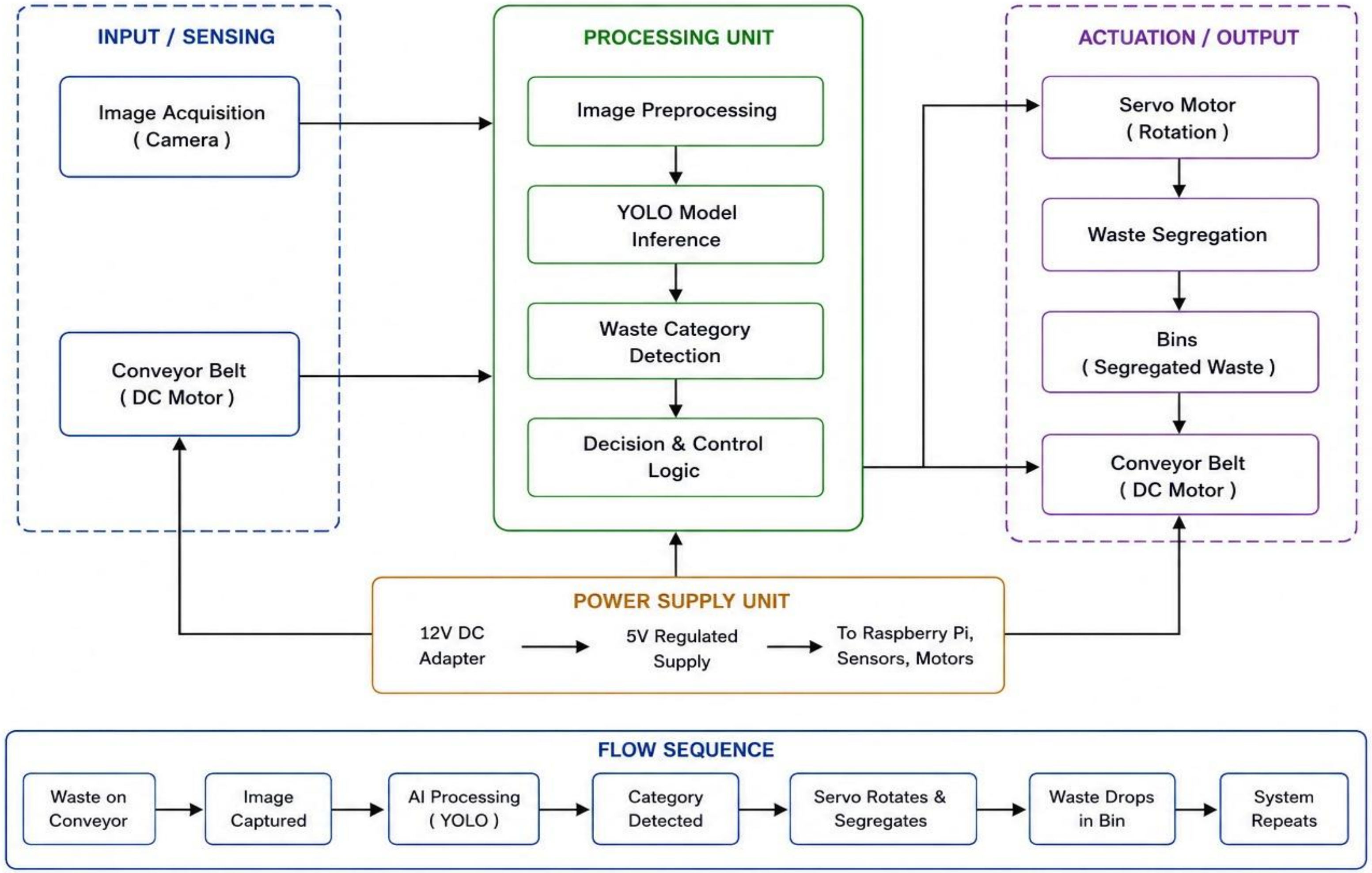
- Continuous rotational motion with 100rpm
- Drives conveyor belt movement
- High torque operation
- Low power consumption
- Simple control mechanism.
- Controls transportation of waste objects

5.Web Camera-

Web Camera W200 Features:

- High-resolution with 1280x720 pixels (720p) image capture
- Frame Rate: Upto 30 frames per second (fps)
- USB interface support
- Field of View: 72-degree wide-angle view
- Focus: fixed focus

AI Based Waste Segregation System – Block Diagram



➢ **Algorithm:**

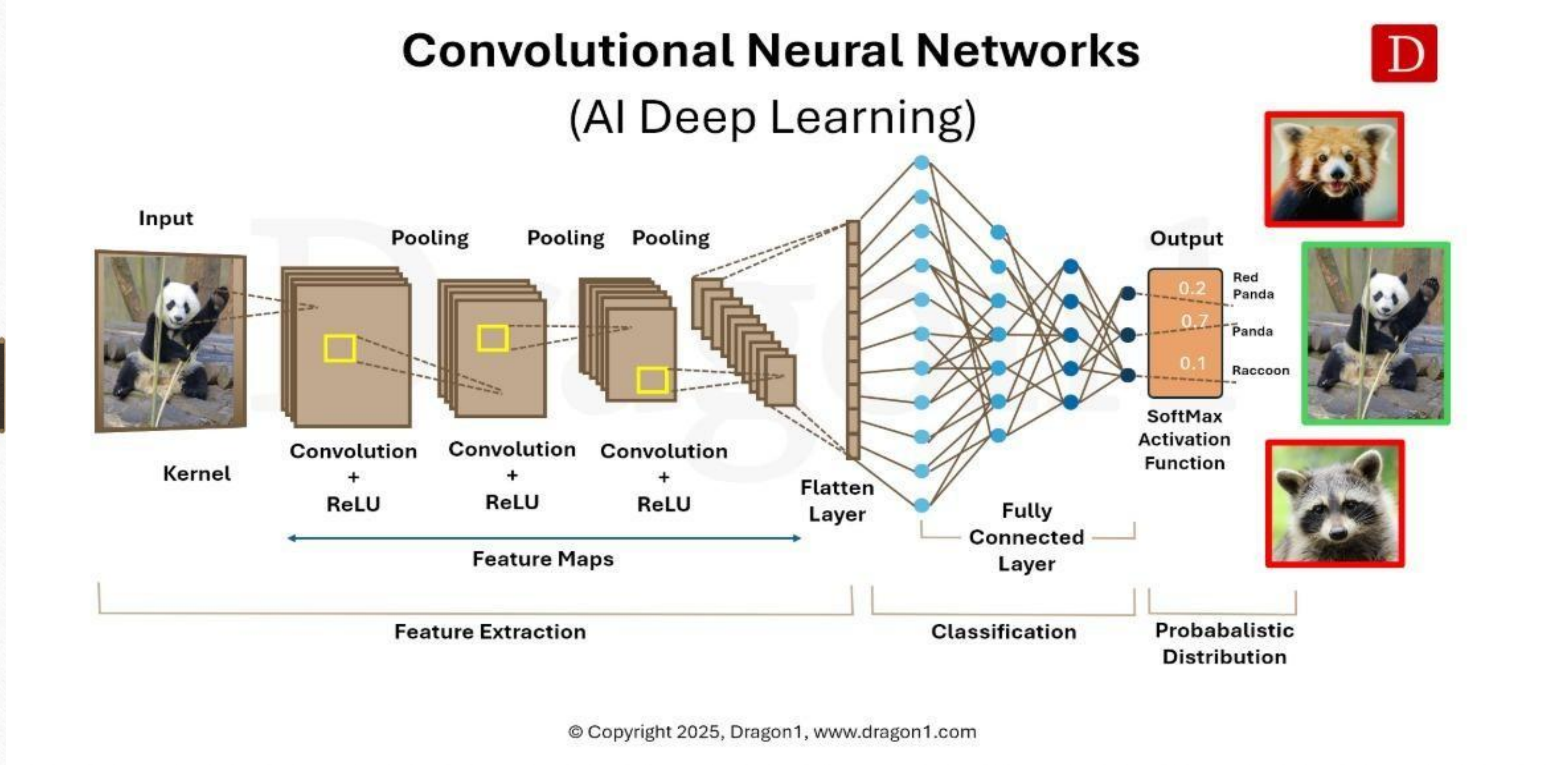
1. Start
2. Initialize camera and GPIO Pins
3. Start Conveyor belt
4. While System is ON:
    a. Stop conveyor belt
    b. Capture image of waste
    c. Preprocess the image
    d. Run CNN model inference
    e. Identify waste category
    f. Activate corresponding servo motor (sorting mechanism)
    g. Wait for fixed delay
    h. Reset servo to default position
    i. Resume conveyor belt
5. End

Software Techniques:

Tool	Description
Dataset	Collection of labelled waste images(plastic, metal, paper, wood etc.) used for training the model
Roboflow	Used for dataset annotation, preprocessing and augmentation to improve model accuracy
YOLO(CNN model)	Real-time object detection model used to identify and classify waste types
TensorFlow	Converts and deploys the trained model(TFLite) for efficient execution on embedded system

➢ **WORKING**

- When the system is powered ON, the conveyor belt starts moving waste
- As waste reaches the detection area, the camera captures its image
- The captured image is processed and adjusted (resized and normalised)
- The processed image is given to the YOLO model for analysis
- The AI model scans the image and identifies the type of waste based on learned features
- The model outputs the waste category with a confidence score
- The system selects the most confident result
- This result is sent to the Raspberry Pi
- The Pi rotates the servo motor to a specific angle
- The waste is divided into the corresponding bin
- After sorting, the servo returns to its original position
- The conveyor resumes movement for the next waste item
- The entire process repeats continuously for automatic segregation


Convolutional Neural Networks
(AI Deep Learning)
Input
Kernel
Pooling
Pooling
Pooling
Convolution
+
ReLU
Convolution
+
ReLU
Convolution
+
ReLU
Feature Maps
Flatten
Layer
Fully
Connected
Layer
Output
0.2
0.7
0.1
Red
Panda
Panda
Raccoon
SoftMax
Activation
Function
Feature Extraction
Classification
Probabalistic
Distribution
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CODE	INFERENCE
`interpreter = tf.lite.Interpreter(model_path="model.tflite")` `interpreter.allocate_tensors()`	• We use a TensorFlow Lite model for lightweight real-time waste classification on Raspberry Pi. allocate_tensors() prepares memory for inference.
`labels = ['cloth', 'metal', 'paper', 'plastic’]`	• These are the output classes trained in our AI model. The model predicts one among these waste categories
`img = cv2.resize(frame, (input_size, input_size))` `img = img.astype(np.float32) / 255.0`	• The captured image is resized and normalized before giving it to the AI model. Normalization improves prediction accuracy.
`interpreter.set_tensor(input_details[0]['index'], img)` `interpreter.invoke()`	• This is the actual AI prediction step. The image is passed into the model and inference is performed.

CODE	INFERENCE

CONFIDENCE FILTERING:
 	
`if obj_conf < 0.25:`
`continue`
	• Low-confidence detections are ignored to reduce false predictions

BEST PREDICTION SELECTION: `class_id = np.argmax(class_scores)`	• The class with the highest probability score is selected as the detected waste type

PREDICTION BUFFER:
 	
`prediction_buffer.append(best_label)`
`BUFFER_SIZE = 7`
	• Instead of acting on a single frame prediction, we store multiple predictions in a buffer to improve stability

RESET MECHANISM: `object_locked = False`	• After one waste object is processed and removed, the system resets itself to detect the next object.

- **Expected Outcomes:**
- Accurate identification and segregation of waste materials
- Reduction in manual sorting effort
- Faster and efficient waste processing
- Improved cleanliness and recycling efficiency
- Low-cost and scalable solution

- **Applications:**
- Smart waste management systems
- Municipal waste segregation plants
- Recycling industries

- **Future Scope:**
- Integration with IoT for real time monitoring
- Use of Advanced AI models for higher accuracy
- Sorting of more waste categories (glass, e-waste, organic and mixed waste)
- Mobile app for system control and analytics
- Implementaion in large-scale industrial setups

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[4]. S. Mittal, R. Sharma, and A. Kumar, "Waste Classification Using Deep Learning Convolutional Neural Networks," International Journal of Computer Applications, vol. 182, no. 38, pp. 1–6, 2019.

[5]. A. Gupta and P. Jain, "Smart Waste Management System Using Machine Learning and IoT," IEEE Access, vol. 8, pp. 124–135, 2020.